

HYDRO ANALYTICS

Historical Data MAF Documentation

Technical Reference — Data API, Station Directory & MAF Calculation Methodology

BASE URL

`http://172.18.7.35:8000/proxy_api_daily`

1. Historical Data API (CSV + DB)

The history API merges two data sources:

- CSV archives (2014-01-01 → 2025-08-18): Pre-loaded historical river data files
- SQLite database (2025-08-19 → present): Live ingested data from FFD PMD

The API finds the intersection of your requested date range with each source, fetches from both, and concatenates the results.

/api/history

Get combined historical inflow/outflow data for a specific station or all stations.

URL

```
GET /proxy_api_daily/api/history
```

Query Parameters

Parameter	Type	Required	Default	Description
<code>name</code>	string	No	all	Station name (see full list, Section 2). If omitted or set to all / * / all_stations, returns ALL stations.
<code>days</code>	integer	No	—	Return last N days of data from now
<code>start_date</code>	string	No	2014-01-01	Start date in YYYY-MM-DD format
<code>end_date</code>	string	No	today	End date in YYYY-MM-DD format

NOTE

Priority: If both `start_date` + `end_date` AND `days` are provided, the date range takes priority. If NEITHER is provided, returns ALL data from 2014 to present.

Example Requests

```
# Kotri, last 7 days
curl "http://172.18.7.35:8000/proxy_api_daily/api/history?name=Kotri&days=7"

# Kotri, monsoon 2026
curl
"http://172.18.7.35:8000/proxy_api_daily/api/history?name=Kotri&start_date=2026-06-01&end_date=2026-09-30"

# Kotri, full history (2014 to now)
curl "http://172.18.7.35:8000/proxy_api_daily/api/history?name=Kotri"
```

```
# Sukkur, last 30 days
curl "http://172.18.7.35:8000/proxy_api_daily/api/history?name=Sukkur&days=30"
```

Response — single station

```
{
  "success": true,
  "name": "Kotri",
  "start_date": "2026-06-01",
  "end_date": "2026-09-07",
  "days": null,
  "inflow": [
    { "x": "2026-06-01T06:00:00", "y": 85000 },
    { "x": "2026-06-02T06:00:00", "y": 92000 }
  ],
  "outflow": [
    { "x": "2026-06-01T06:00:00", "y": 82000 },
    { "x": "2026-06-02T06:00:00", "y": 89000 }
  ],
  "points": 98,
  "source": "csv+database"
}
```

Field	Type	Description
success	boolean	Whether the query succeeded
name	string	Requested station name
start_date	string	Effective start date
end_date	string	Effective end date
days	int / null	Days parameter if provided
inflow	array	Array of { "x": ISO-timestamp, "y": cusecs }
outflow	array	Array of { "x": ISO-timestamp, "y": cusecs }
points	int	Max of inflow/outflow count
source	string	"csv+database" or "database_readonly"

Response — all stations (when name is omitted or all)

```
{
  "success": true,
  "start_date": "2026-06-01",
  "end_date": "2026-09-07",
  "days": 30,
```

```
"total_stations": 24,
"source": "csv+database",
"stations": {
  "KOTRI": { "inflow": [...], "outflow": [...] },
  "SUKKUR": { "inflow": [...], "outflow": [...] },
  "GUDDU": { "inflow": [...], "outflow": [...] },
  "TARBELA": { "inflow": [...], "outflow": [...] }
}
```

Errors

Code	Condition
400	Invalid start_date or end_date format (not YYYY-MM-DD)

/api/history-all

Shortcut to get historical data for ALL stations at once (identical to /api/history?name=all).

URL

```
GET /proxy_api_daily/api/history-all
```

Query Parameters

Parameter	Type	Required	Default	Description
start_date	string	No	2014-01-01	Start date YYYY-MM-DD
end_date	string	No	today	End date YYYY-MM-DD
days	integer	No	—	Return last N days

Examples

```
curl "http://172.18.7.35:8000/proxy_api_daily/api/history-all?days=7"
curl "http://172.18.7.35:8000/proxy_api_daily/api/history-all?start_date=2024-06-01&end_date=2024-09-30"
```

/api/history-csv

Returns ONLY CSV data (no database). For debugging and verifying the CSV archive.

URL

```
GET /proxy_api_daily/api/history-csv
```

Query Parameters

Parameter	Type	Required	Description
name	string	Yes	Station name
start_date	string	Yes	Start date YYYY-MM-DD
end_date	string	Yes	End date YYYY-MM-DD

IMPORTANT
The CSV window is 2014-01-01 to 2025-08-18 only. Dates outside this range are clamped automatically.

Response

```
{
  "success": true,
  "name": "Kotri",
  "start_date": "2024-06-01",
  "end_date": "2024-09-30",
  "csv_window_start": "2014-01-01T00:00:00",
  "csv_window_end": "2025-08-18T23:59:59",
  "inflow": [{ "x": "2024-06-15 06:00", "y": 85000 }],
  "outflow": [{ "x": "2024-06-15 06:00", "y": 82000 }],
  "csv_points": { "inflow": 120, "outflow": 120 }
}
```

Errors

Code	Condition
400	Missing name, start_date, or end_date
400	Invalid date format

2. All Accepted Station Names

Indus River

Station
Tarbela
Attock
Kalabagh
Chashma
Taunsa
Guddu
Sukkur
Kotri

Jhelum River

Station
Kohala
Mangla
Rasul
Kotli

Chenab River

Station
Marala
Khanki
Qadirabad
Chiniot
Trimmu
Panjnad
Jammu Tawi
Akhnur

Ravi River

Station
Jassar
Shahdara
Balloki
Sidhnai

Sutlej River

Station
Sulemanki
Islam
G.S. Wala
Bhakra

Kabul River

Station
Nowshera

Other

Station
Chattar Kalas
Ravi Syphon
Partab Bridge

3. Data Sources & Date Windows

Source	File Location	Date Range	Notes
Combined River CSVs (2014–2024)	<code>data/historical_river_data/combined_river_data_*.csv</code>	2014-01-01 → 2024-12-31	Multiple CSV files, one per year
2025 Flood Data CSV	<code>data/historical_river_data/historic2025floooddata_16june.csv</code>	2025-06-15 → 2025-08-18	Monsoon 2025 data
SQLite Database (live)	<code>data/hydro_history.db</code>	2025-08-19 → present	Continuously ingested via FFD

4. Kotri MAF Calculation

$$MAF_{daily} = Q_{avg} \times T/D$$

5. Quick Reference

```
# Full Kotri history (2014 to now)
curl "http://172.18.7.35:8000/proxy_api_daily/api/history?name=Kotri"

# Kotri monsoon 2026
curl
"http://172.18.7.35:8000/proxy_api_daily/api/history?name=Kotri&start_date=2026-
06-01&end_date=2026-09-30"

# All barrages, last 7 days
curl "http://172.18.7.35:8000/proxy_api_daily/api/history-all?days=7"

# Verify CSV data for Kotri (2024 monsoon)
curl "http://172.18.7.35:8000/proxy_api_daily/api/history-
csv?name=Kotri&start_date=2024-06-01&end_date=2024-09-30"
```